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Recirculating Aquaculture Mixed-Cell Raceway System

Abstract

A recirculating aquaculture Mixed-Cell Raceway (MCR) system enables the implementation of a compact commercial-size fish culturing system in long and narrow agricultural-style buildings commonly used to house and grow animals. The system can be retrofitted to existing agricultural-style buildings or in new constructions. The system is entirely contained within the footprint of a single rectangular MCR. The MCR can be divided into several growth cells equal in width and length. One or more growth cells are dedicated to water quality treatment and pumping. Each growth cell follows diameter-to-depth ratio guidelines previously developed for round tanks to quickly force the removal of solids being generated because of the feeding fish being cultured. All growth cells are in alignment, co-planar, and all the functions are integrated into the MCR's footprint. Minimal disturbance to the building is necessary, and existing utilities are reused so that plumbing, electrical infrastructure, and construction costs are reduced.

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Background/Summary

[0001] The current application claims a priority to the U.S. Provisional Patent application Ser. No. 63/561,713 filed on Mar. 5, 2024.

FIELD OF THE INVENTION

[0002] The present invention generally relates to aquaculture growth systems. More specifically, the present invention discloses an integrated, inline, co-planar mixed-cell raceway fish rearing system that is suitable for retrofit to existing former poultry and swine agricultural-style buildings or industrial buildings.

BACKGROUND OF THE INVENTION

[0003] Many farms in rural United States have long narrow buildings that were built specifically to house and rear poultry and swine. The specifications of these buildings are dictated by the integrated poultry or swine business companies (referred to as integrators) that the farmers choose to enter into a contract to grow animals for the integrator. The integrator also typically supplies the farmers with the animals such as chicks, poults, or piglets, as well as the necessary feed, and then collects the live animals for processing and marketing. Unfortunately, agricultural-style buildings age and become obsolete over time since the integrator often requires the farmer to make upgrades to the buildings. In addition, the farmer may lose the integrator's contract for a variety of reasons. Abandoned buildings frequently have no alternate use and sit by idly to deteriorate with no means of generating income for the farmer due to various circumstances. Moreover, farmers who have installed agricultural-style buildings face the costly upgrades over time, and the ones who do not perform constant upgrades face not receiving new flocks as often as farmers or agricultural-style buildings that are performing at a higher level for the integrator.

[0004] Repurposing agricultural-style buildings can give the farmer a means to generate revenue if the new use falls within the scope and general knowledge base of farming operations. Reusing agricultural-style buildings saves investment costs as the existing buildings provide adequate shelter, already have access to water and electricity, access to semi-skilled labor, and access to farmland for utilization of waste for soil amendment and improvement. All these existing features reduces the cost of investment and improves the profitability and return on invested capital of the farm. For example, an aquaculture growth system could be implemented on the existing unused buildings. The problem in applying an aquaculture growth system to agricultural-style buildings is that conventional aquaculture fish farm systems are too complex and expensive to allow the farmer to operate at an attractive rate of profitability. Current systems have required complex plumbing and expensive water treatment equipment that are difficult to operate and are generally beyond the ability of small-scale farmers to construct and operate profitably. Therefore, a recirculating aquaculture mixed-cell raceway system that can be retrofitted to existing unused agricultural-style buildings in a simplistic manner is necessary and beneficial.

SUMMARY OF THE INVENTION

[0005] The present invention discloses a recirculating aquaculture Mixed-Cell Raceway (MCR) system. The system of the present invention is designed to easily fit into long narrow agricultural-style buildings with minimal disruption to existing floors and structures. The present invention's system is also designed to take advantage of existing water, electricity, waste systems, and other infrastructure already present at the farm site. Further, the present invention's system is designed to collect all functions of water treatment including, but not limited to, Carbon Dioxide (CO₂) stripping, oxygenation, biofiltration, mechanical filtration, and pumping within the envelope of the MCR, specifically within a water treatment section.

[0006] The water treatment section of the present invention is generally at one end of the MCR and is about 25% of the volume of the total MCR (e.g., a 6-growth cell unit would have one or more cells for water treatment), and the water treatment section volume is proportional to the feed loading of the fish growth cells. Further, water preferably moves out of the fish-containing growth cells in two different pathways: [0007] 1) A majority of the waterflow moves under force of gravity from the fish-containing growth cells to the treatment section. The treatment section includes three functional treatment reservoirs fluidly connected to each other through progressively lower weirs until the waterflow reaches lifting pumps in a pumping reservoir. The lifting pumps then re-elevate and distribute the filtered waterflow back to each of the fish-containing growth cells, to repeat this process continually. [0008] 2) A minority of the waterflow moves through center drains of each fish-containing growth cells. Solids are collected at the center drains by the rotation of the waterflow in each fish-containing growth cells. The waterflow then moves through center drain pipes to a side channel or pipe which conveys the waterflow to a solids removal device. The solids removal device collects and removes the solid waste from the water flow. Then, the filtered waterflow is expelled into the pumping reservoir where the filtered waterflow can be reunited with the majority of the waterflow.

Further, the treatment section is preferably located at one end of the MCR; however, the water treatment section or additional treatment sections can be located at any position along the MCR. For example, the water treatment section can be positioned among the fish-containing growth cells. The particular chosen position can be decided based on the needs of the fish species, the specific husbandry process of the farming operation, the specifics of the design of the existing barn, or the bias of the farmer operator.

[0009] Further, the present invention's system can be designed to house a single cohort of fish (no barriers between the virtual cells) or segregate and house multiple cohorts (ages) of fish simultaneously. Several water dividing gates between the cohorts can be utilized to segregate the multiple cohorts and to hydraulically isolate the final stage (pre-shipment) of growth just before delivery to harvesting in a process known as purging. Purging is the process where common system water is excluded from one or more growth cells and new water is added to remove any off flavors the fish may possess. As handling fish causes the fish to stress and to slow the growth, the present invention's system allows the farmer to never have to handle the fish during their growth period and subsequent purging period, saving the farmer's labor and time and minimizing stress to the fish.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a top perspective view of the Mixed-Cell Raceway (MCR) of the system of the present invention, wherein the aquaculture growth section is shown with several dividing gates and a purging gate.

[0011] FIG. 2 is a magnified view of the water treatment section of the MCR of the present invention's system shown in FIG. 1.

[0012] FIG. 3 is a magnified view of the aquaculture growth section of the MCR of the present invention's system shown in FIG. 1.

[0013] FIG. 4 is a top schematic view of the MCR of the present invention's system, wherein the aquaculture growth section is shown without the dividing gates nor the purging gate.

[0014] FIG. 5 is a vertical cross-sectional view of the MCR of the present invention's system taken along line 5-5 shown in FIG. 4.

[0015] FIG. 6 is a vertical cross-sectional view of the MCR of the present invention's system taken along line 6-6 shown in FIG. 4.

[0016] FIG. 7 is a vertical cross-sectional view of the MCR of the present invention's system taken along line 7-7 shown in FIG. 4.

[0017] FIG. 8 is a magnified view of the water treatment section of the MCR of the present invention's system shown in FIG. 4.

[0018] FIG. 9 is a magnified view of the aquaculture growth section of the MCR of the present invention's system shown in FIG. 4.

[0019] FIG. 10 is a magnified view of the water treatment section of the MCR of the present invention's system shown in FIG. 5.

[0020] FIG. 11 is a top perspective view of the Moving Bed Biofilter Reactor (MBBR) reservoir of the present invention's system.

[0021] FIG. 12 is a front perspective view of a floating surface aerator of the MBBR reservoir of the present invention's system.

[0022] FIG. 13 is a magnified top perspective view of an alternate embodiment of the water treatment section of the present invention's system, wherein the water treatment section is shown with two Static Bed Filter (SBF) reservoirs.

[0023] FIG. 14 is a top perspective view of an alternate embodiment of the MCR of the present invention's system, wherein the four aquaculture growth sections are shown connected to the water treatment section, and wherein multiple SBF reservoirs and several pump reservoirs are shown in the water treatment section.

[0024] FIG. 15 is a top perspective view of an alternate embodiment of the MCR of the present invention's system, wherein the four aquaculture growth sections are shown connected to the water treatment section, and wherein multiple SBF reservoirs and several pump reservoirs are shown in the water treatment section.

[0025] FIG. 16 is a top perspective view of an alternate embodiment of the MCR of the present invention's system, wherein the four aquaculture growth sections are shown connected to the water treatment section, and wherein several SBF reservoirs and several pump reservoirs are shown in the water treatment section.

[0026] FIG. 17 is a top perspective view of the MCR of the present invention's system, wherein an alternate arrangement of the water treatment section is shown.

[0027] FIG. 18 is a top perspective view of the MCR of the present invention's system, wherein the alternate arrangement of the water treatment section is shown.

[0028] FIG. 19 is a top perspective view of the MCR of the present invention's system, wherein the alternate arrangement of the water treatment section is shown.

[0029] FIG. 20 is a magnified top perspective view of the alternate arrangement of the water treatment section of the present invention's system.

[0030] FIG. 21 is a top perspective view of a long and narrow agricultural-style building that can be retrofitted with the present invention's system.

DETAIL DESCRIPTIONS OF THE INVENTION

[0031] All illustrations of the drawings are for the purpose of describing selected versions of the present invention and are not intended to limit the scope of the present invention.

[0032] The present invention discloses a recirculating aquaculture Mixed-Cell Raceway (MCR) system. The present invention's system enables the implementation of a simplified, inline,

recirculating, MCR in long and narrow agricultural-style buildings with minimal disruption to the existing building's floors and structure, as can be seen in FIG. 21. In addition, the present invention's system allows the utilization of existing water, electricity, and waste systems found in the target building. The present invention's system is not restricted to existing agricultural-style buildings, other suitable existing or newly constructed buildings can be utilized. As can be seen in FIGS. 1, 4, and 17 through 20, in the preferred embodiment, the present invention comprises at least one MCR 1 and a drain system 34, as can be seen in FIG. 1 through 7. The at least one MCR 1 corresponds to an integrated structure that houses all subsystems of the present invention's system that facilitate the growth of the target aquatic organisms. The at least one MCR 1 also enables the necessary flow through the system that facilitates the filtration of the waterflow. Further, the drain system 34 enables secondary flow that removes solid waste from the waterflow.

[0033] The general configuration of the aforementioned components enables farmers to repurpose unused buildings to grow different aquatic organisms in a cost-effective manner. As can be seen in FIGS. 1, 4, and 13 through 20, the at least one MCR 1 is designed to house all necessary subsystems that facilitate the growth of the target aquatic organisms with minimal handling. As can be seen in FIG. 4 through 10, the at least one MCR 1 comprises at least one aquaculture growth section 2, a water treatment section 14, and a water return system 30. The at least one aquaculture growth section 2 corresponds to the section of the at least one MCR 1 where the target aquatic organisms grow. The water treatment section 14 corresponds to the section that performs several water treatment processes to ensure the healthy growth of the target aquatic organisms. So, the water treatment section 14 comprises a first Static Bed Filter (SBF) reservoir 15, a Moving bed biofilter reactor (MBBR) reservoir 23, and a pumping reservoir 26. The first SBF reservoir 15 enables the removal of suspended particles and contaminants from the waterflow. The MBBR reservoir 23 enables the performance of different water treatment functions including, but not limited to, oxygenation, Carbon Dioxide (CO₂) removal, organic carbon removal, nitrification, etc. The pumping reservoir 26 corresponds to the structure that temporarily retains the treated water that is moved back to the at least one aquaculture growth section 2 via the water return system 30. The water return system 30 enables the flow of the treated water from the water treatment section 14 to the at least one aquaculture growth section 2 to form a closed continuous waterflow loop.

[0034] In the preferred embodiment, the present invention can be implemented as follows: the at least one aquaculture growth section 2 is positioned adjacent to the water treatment section 14 to maintain both sections contained within the at least one MCR 1, as can be seen in FIGS. 1, 4, and 13 through 20. In addition, the at least one aquaculture growth section 2 and the water treatment section 14 are positioned coplanar with each other to eliminate the need for modifications to the existing building's floor. Further, the at least one aquaculture growth section 2, the first SBF reservoir 15, the MBBR reservoir 23, and the pumping reservoir 26 are in fluid communication with each other so that waterflow is enabled through the different sections of the at least one MCR 1. In other words, water can flow from the at least one aquaculture growth section 2, through the first SBF reservoir 15 and the MBBR reservoir 23, and into the pumping reservoir 26. Further, the at least one aquaculture growth section 2 is in fluid communication with the pumping reservoir 26 via the drain system 34 so a secondary waterflow pathway is formed from the at least one aquaculture growth section 2 to the pumping reservoir 26. Furthermore, the pumping reservoir 26 is in fluid communication with the at least one aquaculture growth section 2 via the water return system 30 to enable the return of treated water back into the at least one aquaculture growth section 2. Thus, a closed waterflow loop is established that enables the continuous treatment of water in the at least one MCR 1 to promote the healthy growth of the target aquatic organisms.

[0035] As previously discussed, the drain system 34 enables the flow of untreated water from the at least one aquaculture growth section 2 to the pumping reservoir 26. In addition, the drain system 34 enables the removal of solid waste from the untreated water before the waterflow moves into the pumping reservoir 26. As can be seen in FIG. 1 through 10 and 13 through 20, the at least one

aquaculture growth section **2** may comprise at least one growth cell **3**. The at least one growth cell **3** corresponds to a subdivision inside the at least one aquaculture growth section **2** that retains the target aquatic organism during the growth phase. The at least one growth cell **3** is preferably a rectangular shaped structure with an open top. So, the at least one growth cell **3** generally comprises a cell opening **4** and a cell floor **5**. In addition, the drain system **34** comprises at least one gravity-assisted drain **35**, a drain conduit **36**, and a filtration reservoir **37**. The at least one gravity-assisted drain **35** corresponds to a structure on the at least one growth cell **3** that allows the draining of the untreated water via gravity towards the water treatment section **14**. The drain conduit **36** corresponds to a flow mechanism that allows the flow of untreated water from the at least one growth cell **3** to the filtration reservoir **37**. The filtration reservoir **37** is designed to remove solid waste from the untreated waterflow before moving into the pumping reservoir **26**.

[0036] In the preferred embodiment, the drain system **34** can be implemented as follows: the drain conduit **36** is positioned external to the at least one mixed-cell raceway, as can be seen in FIG. **1** through **10** and **13** through **20**. The drain conduit **36** is preferably a structure separate from the at least one MCR **1**. For example, the drain conduit **36** can be an elongated channel that spans the length of the at least one MCR **1** to guide the untreated waterflow from the at least one growth cell **3** to the pumping reservoir **26**. Further, the at least one gravity-assisted drain **35** is centered on the cell floor **5**. Since the flow of water in the at least one growth cell **3** is preferably rotating in the clockwise or counterclockwise direction, solid waste is guided towards the center of the cell floor **5** so that the solid waste is collected by the at least one gravity-assisted drain **35**. Further, the at least one gravity-assisted drain **35** is integrated into the cell floor **5** to secure the at least one gravity-assisted drain **35** to the cell floor **5**. The at least one gravity-assisted drain **35** is also in fluid communication with the filtration reservoir **37** via the drain conduit **36**. For example, the at least one gravity-assisted drain **35** can be implemented on a drain hole on the cell floor **5** and is designed to enable the flow of water and small solid waste out of the at least one growth cell **3**. The at least one gravity-assisted drain **35** also prevents the growing aquatic organisms from exiting the at least one growth cell **3**. Further, the at least one gravity-assisted drain **35** can be connected to the drain conduit **36** via the appropriate tubing that is installed below the cell floor **5**. Alternatively, the at least one gravity-assisted drain **35** can be provided as a pipe installed above the cell floor **5** which would syphon out water from the center drain area. A syphon-type drain is often used so that no holes are made in the floor of the at least one MCR **1**. Furthermore, the filtration reservoir **37** is in fluid communication with the pumping reservoir **26** to complete the fluid connection. Thus, untreated water with solid waste can exit the at least one growth cell **3** through the at least one gravity-assisted drain **35** into the drain conduit **36**. The untreated water then moves through the drain conduit **36** towards the filtration reservoir **37** where solid waste is removed from the untreated water. As a result, the treated water is moved into the pumping reservoir **26** so that the treated water can be moved back into the at least one growth cell **3**.

[0037] As previously discussed, the drain system **34** is preferably implemented external to the at least one MCR **1**. This can be done by utilizing the appropriate tubing or similar fluid mechanisms that connect the drain conduit **36** to the at least one gravity-assisted drain **35** and the filtration reservoir **37** to the pumping reservoir **26**. However, in some embodiments, the filtration reservoir **37** is mounted within the pumping reservoir **26** so that the filtration reservoir **37** is in direct fluid communication with the pumping reservoir **26**, as can be seen in FIG. **1** through **10** and **13** through **20**. For example, the filtration reservoir **37** can be implemented so that the top of the filtration reservoir **37** is open, which enables the treated water to flow out of the filtration reservoir **37** and into the pumping reservoir **26**. In other embodiments, different fluid mechanisms can be implemented to enable the passive outflow of treated water from the filtration reservoir **37** into the pumping reservoir **26**.

[0038] As can be seen in FIG. **1** through **10** and **13** through **20**, to help remove the solid waste from the untreated water flowing through the filtration reservoir **37**, the drain system **34** may further

comprise a solids removal device **38**. The solids removal device **38** can be a rotary drum filter that allows the removal of solid waste from the untreated water without blocking the waterflow through the filtration reservoir **37**. So, in general, the solids removal device **38** is mounted within the filtration reservoir **37** to secure the solids removal device **38** within the filtration reservoir **37**. Further, the filtration reservoir **37** is in fluid communication with the drain conduit **36** via the solids removal device **38** so that the untreated water passes through the solids removal device **38** while flowing through the filtration reservoir **37**. In other embodiments, different filtration mechanisms can be implemented that remove solid waste from the untreated waterflow.

[0039] While the at least one MCR **1** includes at least one growth cell **3**, the present invention's system preferably includes at least one MCR **1** with a plurality of growth cells **6**, as can be seen in FIG. **1** through **10** and **13** through **20**. Each of the plurality of growth cells **6** can be accommodated to retain different aquatic organisms so that all the aquatic organisms can grow simultaneously in the at least one aquaculture growth section **2**. In addition, the at least one gravity-assisted drain **35** is also a plurality of gravity-assisted drains **39** so that each growth cell of the plurality of growth cells **6** has a corresponding gravity-assisted drain. Further, the plurality of growth cells **6** is distributed along the at least one aquaculture growth section **2** so that the several growth cells are evenly distributed throughout the at least one aquaculture growth section **2**. Furthermore, each gravity-assisted drain of the plurality of growth cells **6** is integrated into the closed base of the corresponding growth cell of the plurality of growth cells **6**. The untreated water from each growth cell can flow into the drain conduit **36** to maintain all the plurality of growth cells **6** in optimal conditions for the healthy growth of the retained aquatic organisms. Thus, a minority of the waterflow exiting the at least one aquaculture growth section **2** through force of gravity carries away most of the large solid waste from each of the plurality of growth cells **6**. In other embodiments, the drain conduit **36** is not limited to being configured as an open channel; the drain conduit **36** can also be configured as a semi-closed or closed pipe.

[0040] As previously discussed, the waterflow inside each growth cell is preferably arranged to rotate counter to the rotational direction of each adjacent growth cell. This is achieved by guiding the flow of the treated water returning from the pumping reservoir **26** horizontally into the growth cell to drive counterclockwise or clockwise flow depending upon the rotational direction of adjacent cells of the water inside the growth cell. As can be seen in FIG. **1** through **10** and **13** through **20**, the water return system **30** may further comprise at least one fluid pump **31** and at least one water return manifold **32**. The at least one fluid pump **31** helps move the treated water from the pumping reservoir **26** back into the at least one growth cell **3**. The at least one water return manifold **32** guides the outflow of the returning treated water horizontally into the at least one growth cell **3**.

[0041] In general, the at least one water return manifold **32** can be implemented as follows: the at least one water return manifold **32** is positioned adjacent to the cell opening **4** or along the lateral side so that the treated water flows into the corresponding growth cell, as can be seen in FIG. **1** through **10** and **13** through **20**. The at least one water return manifold **32** is also perimetrically connected to the at least one growth cell **3**. In other words, the at least one water return manifold **32** is mounted onto a lateral wall of the at least one growth cell **3**. The at least one water return manifold **32** is also oriented towards the water treatment section **14** or away from the water treatment section **14** depending on the lateral wall that the at least one water return manifold **32** is mounted on. This ensures that the waterflow within the at least one growth cell **3** rotates counterclockwise or clockwise depending upon how many growth cells are present and the rotational condition of adjacent cells. On the other hand, the at least one fluid pump **31** is mounted within the pumping reservoir **26** so that the at least one fluid pump **31** can move the treated water out of the pumping reservoir **26**. Further, the at least one water return manifold **32** is in fluid communication with the pumping reservoir **26** via the at least one fluid pump **31**. This way, as the treated water is pumped back into the at least one growth cell **3** through the at least one water

return manifold **32**, the water outflow is horizontal and close to the outer perimeter of the at least one growth cell **3**. The waterflow within the at least one growth cell **3** is thus forced to rotate counterclockwise or clockwise depending upon rotation of adjacent cells to facilitate the flow of solid waste downstream towards the at least one gravity-assisted drain **35**.

[0042] When several growth cells are implemented, the water return system **30** includes a plurality of water return manifolds **33** for the plurality of growth cells **6**, as can be seen in FIG. **1** through **10** and **13** through **20**. Similar to the plurality of gravity-assisted drains **39**, each water return manifold of the plurality of water return manifolds **33** is perimetrically connected to the corresponding growth cell of the plurality of growth cells **6**. Thus, treated water is returned to all the plurality of growth cells **6** while ensuring that the waterflow within each growth cell rotates. The waterflow rotate within each growth cell about an axis oriented vertically above the corresponding gravity-assisted drain. Further, an arbitrary growth cell rotates in the opposite direction to the adjacent growth cells. In other words, a growth cell rotates clockwise, and the adjacent growth cells rotate counterclockwise. In other embodiments, different fluid mechanisms can be implemented to accomplish the counterclockwise and/or clockwise rotational waterflow within each of the plurality of growth cells **6**.

[0043] As previously discussed, the first SBF reservoir **15** enables the removal of suspended particles and contaminants from the waterflow. As can be seen in FIG. **1** through **10** and **13** through **20**, the first SBF reservoir **15** may comprise a cell-to-SBF weir **16**, an SBF-to-MBBR weir **17**, a reservoir opening **18**, and a reservoir floor **19**. Like the at least one aquaculture growth section **2**, the first SBF reservoir **15** is a rectangular structure with an open top. The cell-to-SBF weir **16** enables the flow of untreated water from the at least one growth cell **3** to the first SBF reservoir **15** via gravity. The SBF-to-MBBR weir **17** enables the flow of partially treated water from the first SBF reservoir **15** to the MBBR reservoir **23**.

[0044] In general, the first SBF reservoir **15** can be implemented as follows: the cell-to-SBF weir **16** and the SBF-to-MBBR weir **17** are perimetrically distributed about the first SBF reservoir **15**, as can be seen in FIG. **1** through **10** and **13** through **20**. The cell-to-SBF weir **16** is positioned adjacent to the at least one growth cell **3**, while the SBF-to-MBBR weir **17** is positioned adjacent to the MBBR reservoir **23**. Further, the cell-to-SBF weir **16** and the SBF-to-MBBR weir **17** is integrated into the reservoir opening **18** of the first SBF reservoir **15**. In other words, the cell-to-SBF weir **16** is implemented on the wall between the first SBF reservoir **15** and the at least one growth cell **3**, while the SBF-to-MBBR weir **17** is implemented on the wall between the first SBF reservoir **15** and the at least one growth cell **3**. Further, the at least one growth cell **3** is in fluid communication with the first SBF reservoir **15** via the cell-to-SBF weir **16**, while the first SBF reservoir **15** is in fluid communication with the MBBR reservoir **23** via the SBF-to-MBBR weir **17**. The cell-to-SBF weir **16** is also in a gravitational-higher position than the SBF-to-MBBR weir **17** so that the waterflow moves from the at least one growth cell **3**, through the first SBF reservoir **15**, and into the MBBR reservoir **23**. Further, the at least one aquaculture growth section **2** can be physically connected to the water treatment section **14**. The at least one aquaculture growth section **2** can also be physically separated from the water treatment section **14**, and the water treatment section **14** may be constructed from a different material from the at least one aquaculture growth section **2**. In this case, several tubing, piping, or other appropriate fluid mechanisms can be utilized to connect the at least one growth cell **3** to the cell-to-SBF weir **16** so that untreated water from the at least one growth cell **3** can flow into the first SBF reservoir **15**.

[0045] When a plurality of growth cells **6** are implemented in the at least one aquaculture growth section **2**, the plurality of growth cells **6** is arranged so that a majority of the waterflow moving through the at least one aquaculture growth section **2** flows through a single growth cell before moving into the first SBF reservoir **15**, as can be seen in FIG. **1** through **10** and **13** through **20**. In this embodiment, the plurality of growth cells **6** may comprise a first terminal cell **7** corresponding to the growth cell that is in direct fluid communication with the first SBF reservoir **15**. So, the first

terminal cell **7** is positioned adjacent to the water treatment section **14** while the rest of the growth cells are positioned opposite the water treatment section **14** across the first terminal cell **7**. Further, each of the plurality of growth cells **6** is in fluid communication with each other so that waterflow can freely move through the at least one aquaculture growth section **2**. Furthermore, the first terminal cell **7** is in fluid communication with the first SBF reservoir **15** via the cell-to-SBF weir **16**. This way, the majority of the waterflow from the at least one aquaculture growth section **2** moves into the first SBF reservoir **15** through the first terminal cell **7**.

[0046] As can be seen in FIG. **1** through **10** and **13** through **20**, the first SBF reservoir **15** can include different means to passively filtrate the untreated waterflow moving through the first SBF reservoir **15** from the at least one aquaculture growth section **2**. In general, the first SBF reservoir **15** may further comprise a SBF baffle **20**, a SBF media section **21**, and a SBF media barrier **22**. The SBF baffle **20** is a stationary element within the first SBF reservoir **15** that controls the incoming waterflow from the cell-to-SBF weir **16**. The SBF media barrier **22** maintains the filtering media suspended within the first SBF reservoir **15**. The SBF media section **21** corresponds to the area where the filtering media is present inside the first SBF reservoir **15**. So, the SBF baffle **20** is mounted within the first SBF reservoir **15**, adjacent to the cell-to-SBF weir **16**, to force the incoming waterflow from the cell-to-SBF weir **16** towards the SBF media section **21**. The SBF media barrier **22** is mounted within the first SBF reservoir **15**, adjacent to the reservoir opening **18** of the first SBF reservoir **15**. The SBF media barrier **22** can be positioned parallel to the reservoir floor **19** of the first SBF reservoir **15** to increase the area of the SBF media section **21**. In addition, the SBF media section **21** is positioned in between the SBF media barrier **22** and the SBF baffle **20** and can be located above the SBF media barrier **22** for floating media or below the SBF media barrier **22** for neutral or sinking media. The filtration media in the SBF media section **21** filters particles of waste solids from the waterflow before the water upwells through the SBF media barrier **22** which is suspended below the operating level of the SBF water. This way, as the untreated water flows into the first SBF reservoir **15** through the cell-to-SBF weir **16**, the untreated water is forced down into the SBF media section **21** by the SBF baffle **20**. The waterflow then moves through the SBF media section **21** and passes through the SBF media barrier **22**. Then, the partially treated water then moves towards the SBF-to-MBBR weir **17** so that the partially treated water moves into the MBBR reservoir **23**.

[0047] In other embodiments, the first SBF reservoir **15** can alternatively be located downstream of the MBBR reservoir **23**, as can be seen in FIG. **13** through **20**. There are various advantages and disadvantages of placing the first SBF reservoir **15** in either location, and disadvantages could be minimized by implementing two SBF reservoirs before and after the MBBR reservoir **23**. So, in this embodiment, the water treatment section **14** may further comprise a second SBF reservoir **27**. The second SBF reservoir **27** corresponds to the SBF reservoir positioned in between the MBBR reservoir **23** and the pumping reservoir **26**. The second SBF reservoir **27** can include the same filtration features as the first SBF reservoir **15**. In addition, the second SBF reservoir **27** is implemented so that the waterflow through the water treatment section **14** is continuous and uninterrupted. So, the at least one aquaculture growth section **2**, the first SBF reservoir **15**, the MBBR reservoir **23**, the second SBF reservoir **27**, and the pumping reservoir **26** are in fluid communication with each other.

[0048] Like the first SBF reservoir **15**, the second SBF reservoir **27** includes several weirs that allow the waterflow through the second SBF reservoir **27** via gravity. As can be seen in FIG. **13** through **20**, the second SBF reservoir **27** may comprise a MBBR-to-SBF weir **28**, an SBF-to-pump weir **29**, a reservoir opening **18**, and a reservoir floor **19**. Like the weirs of the first SBF reservoir **15**, the MBBR-to-SBF weir **28** and the SBF-to-pump weir **29** are perimetrically distributed about the second SBF reservoir **27**. The MBBR-to-SBF weir **28** is positioned adjacent to the MBBR reservoir **23**, while the SBF-to-pump weir **29** is positioned adjacent to the pumping reservoir **26**. Further, the MBBR-to-SBF weir **28** and the SBF-to-pump weir **29** are integrated into the reservoir

opening **18** of the second SBF reservoir **27** to enable the waterflow through the second SBF reservoir **27**. This way, the MBBR reservoir **23** is in fluid communication with the second SBF reservoir **27** via the MBBR-to-SBF weir **28**. The MBBR-to-SBF weir **28** is in a gravitational-lower position than the SBF-to-MBBR weir **17** so that waterflow moves from the first SBF reservoir **15**, through the MBBR reservoir **23**, and into the second SBF reservoir **27**. Further, the second SBF reservoir **27** is in fluid communication with the pumping reservoir **26** via the SBF-to-pump weir **29**. This way, the waterflow through the second SBF reservoir **27** can continue moving towards the pumping reservoir **26**. In other embodiments, additional SBF reservoirs or different arrangements of the SBF reservoirs can be implemented in the water treatment section **14** as necessary.

[0049] Like the first SBF reservoir **15**, the MBBR reservoir **23** is also preferably a rectangular subsection of the water treatment section **14** with an open top. As can be seen in FIG. **1** through **12**, in general, the MBBR reservoir **23** comprises a MBBR-to-pump weir **24** and a reservoir opening **18**. The MBBR-to-pump weir **24** corresponds to the weir that is implemented when the waterflow moves directly from the MBBR reservoir **23** to the pumping reservoir **26**. Like the weirs of the first SBF reservoir **15**, the MBBR-to-pump weir **24** is perimetrically distributed about the MBBR reservoir **23**. The MBBR-to-pump weir **24** is also positioned adjacent to the pumping reservoir **26** and is integrated into the reservoir opening **18** of the MBBR reservoir **23**. This allows waterflow from the MBBR reservoir **23** to the pumping reservoir **26**. Furthermore, the pumping reservoir **26** is in fluid communication with the MBBR reservoir **23** via the MBBR-to-pump weir **24**. The MBBR-to-pump weir **24** is in a gravitational-lower position than the SBF-to-MBBR weir **17** so that the waterflow through the water treatment section **14** is promoted via gravity. In other embodiments, different fluid mechanisms can be implemented to aid in the waterflow through the water treatment section **14**.

[0050] As can be seen in FIG. **1** through **12**, to further remove waste from the partially treated water flowing from the first SBF reservoir **15**, the MBBR reservoir **23** may further comprise a plurality of surface aerators **25**. The plurality of surface aerators **25** can include, but is not limited to, floating aerators, non-floating aerators, or a combination thereof. The plurality of surface aerators **25** is distributed throughout the MBBR reservoir **23** to maximize the action of the plurality of surface aerators **25** within the MBBR reservoir **23**. Further, the plurality of surface aerators **25** is mounted within the MBBR reservoir **23** so that each surface aerator can remain on the surface as the waterflow moves through the MBB reservoir. This way, when the partially treated water enters the MBBR reservoir **23**, the waterflow is thoroughly mixed with MBBR media by action of the plurality of surface aerators **25**. The energetic mixing of the water, MBBR media, and atmospheric air serves to strip CO₂ from the system water, add Oxygen (O₂) to the waterflow until the water approaches atmospheric saturation, and allows bacteria on MBBR media to have superior contact with the ammonia dissolved in the water. Thus, the dissolved ammonia is changed to a non-harmful nitrogen form. Further, the vigorous mixing of the MBBR media caused by the plurality of surface aerators **25** everywhere also serves to scour the MBBR media on a continuous basis to minimize excessive biofilm accumulation on the MBBR media, which reduces the functionality of the MBBR reservoir **23** in general.

[0051] In the preferred embodiment, the number of the plurality of surface aerators **25** is selected based on the maximum amount of O₂ needed in the present invention's system by the particular aquatic organism being cultured, density, and maximum system loading. Further, most of the O₂ required for culturing the aquatic organism comes from the action of the plurality of surface aerators **25** and supplemental O₂ is only supplied in times of emergency or high loading of the present invention's system. In addition, the water recirculation movement action of the plurality of surface aerators **25** acts to be the force which keeps the MBBR media in constant motion rather than using previous designs of air lifts or other means. The particular use of the plurality of surface aerators **25** and the movement of MBBR media created thereby eliminates one of the major problems with floating media where the media tends to climb out of the water column making this

media unavailable for nitrification.

[0052] Due to the elongated design of the existing agricultural-style building, the at least one MCR **1** is designed as an elongated tank structure to match the elongated design of the existing agricultural-style building. As can be seen in FIG. **1** through **10** and **13** through **20**, to match the elongated shape of the at least one MCR **1**, the plurality of growth cells **6** is preferably implemented in a linear arrangement. In this embodiment, the plurality of growth cells **6** comprises a first terminal cell **7**, a second terminal cell **8**, and a plurality of intermediate cells **9**. The first terminal cell **7** preferably corresponds to the growth cell adjacent to the water treatment section **14**, while the second terminal cell **8** corresponds to the opposite terminal growth cell. The plurality of intermediate cells **9** corresponds to the growth cells positioned between the first terminal cell **7** and the second terminal cell **8**. So, in general, the plurality of growth cells **6** is distributed along the at least one aquaculture growth section **2** due to the elongated design of the at least one MCR **1**. The first terminal cell **7** is positioned adjacent to the water treatment section **14**, while the second terminal cell **8** is positioned opposite the first terminal cell **7** along the at least one aquaculture growth section **2**. Furthermore, the plurality of intermediate cells **9** is positioned in between the first terminal cell **7** and the second terminal cell **8**. As a result, the plurality of growth cells **6** is configured into a linear arrangement along the at least one aquaculture growth section **2**.

[0053] This linear arrangement of the plurality of growth cells **6** allows the at least one aquaculture growth section **2** to retain groups of aquatic organisms separate from each other. For example, different cohort ages of fish or groups of fish can be kept separate from each other during the growing phase. As can be seen in FIG. **1** through **3**, the at least one aquaculture growth section **2** may further comprise a plurality of dividing gates **10** that keep the various cohort ages of fish or groups of fish for no particular reason separated from each other. Each of the plurality of dividing gates **10** span a width of the at least one aquaculture growth section **2** to separate adjacent growth cells. Further, each of the plurality of dividing gates **10** can be selectively mounted within the at least one aquaculture growth section **2** to divide each of the growth cells. For example, a dividing gate of the plurality of dividing gates **10** can be positioned in between the first terminal cell **7** and an adjacent intermediate cell of the plurality of intermediate cells **9**, which divides the first terminal cell **7** from the rest of the growth cells.

[0054] Likewise, a dividing gate of the plurality of dividing gates **10** can be positioned in between the second terminal cell **8** and an adjacent intermediate cell of the plurality of intermediate cells **9** to divide the second terminal cell **8** from the rest of the growth cells. Further, a dividing gate of the plurality of dividing gates **10** can be positioned in between an arbitrary intermediate cell and an adjacent intermediate cell of the plurality of intermediate cells **9**. This allows each intermediate cell from the adjacent intermediate cells to also be divided if necessary. Furthermore, each of the plurality of dividing gates **10** is slidably mounted within the at least one aquaculture growth section **2** so that each dividing gate can be moved, if necessary, without removing the selected dividing gate from the at least one aquaculture growth section **2**.

[0055] As can be seen in FIG. **1** through **3**, to allow waterflow throughout the at least one aquaculture growth section **2**, each dividing gate of the plurality of dividing gates **10** is made of a water permeable material. This allows waterflow through the plurality of dividing gates **10** while retaining the desired aquatic organisms within the target growth cell. Further, each of the plurality of dividing gates **10** is designed to allow the managing of the growing aquatic organisms without overhandling. For example, the youngest and smallest fish in a cohort can be arranged to occupy a single growth cell using the corresponding dividing gates. After an appropriate time and amount of growth, these fish are moved to the unoccupied, adjacent growth cells via an opening in the corresponding dividing gates to allow additional growth. After a similar appropriate length of time, the same cohort of fish are allowed to pass through the remaining dividing gates via the corresponding openings to finish growing in the remaining unoccupied growth cells. Further, each of the plurality of dividing gates **10** can be fixed in place or be moved along the at least one

aquaculture growth section **2** to encourage the fish to move into the adjoining growth cell when appropriate. These fish movements all happen without ever touching the fish, which reduces stress on the fish and promotes faster growth while reducing labor needed in the operation of the farm. Alternatively, the at least one aquaculture growth section **2** can be arranged to house a single cohort of fish without the use of the plurality of dividing gates **10**.

[0056] In some embodiments, the at least one aquaculture growth section **2** may further comprise an inlet water manifold **11** and a purging gate **12**, as can be seen in FIG. **1** through **3**. The purging gate **12** is a minimally water permeable barrier that restricts new water released from the inlet water manifold **11** from mixing with recirculating system water. The purging gate **12** also allows a predetermined quantity of harvest-size fish to enter the new water area via a door in the purging gate **12** to be immersed only in new water. This reduces any off-flavors remaining in the harvest-size fish before the fish are harvested in a process called purging. The purging process flushes enzymes, off-flavor-producing bacteria, and other byproducts so that the harvest-ready fish are ready for market harvest. The purging process also keeps these fish separate from the fish that are still being actively grown. Furthermore, the purging gate **12** comprises an outlet weir **13** that allows the flow of fresh water to flow into the rest of the plurality of growth cells **6** via gravity.

[0057] As can be seen in FIG. **1** through **3**, the purging gate **12** and the inlet water manifold **11** are preferably implemented as follows: the inlet water manifold **11** is positioned adjacent to the second terminal cell **8** so that fresh water flows into the present invention's system from the second terminal cell **8**. In addition, the inlet water manifold **11** is mounted onto a cell opening **4** of the second terminal cell **8** to allow the flow of fresh water down into the second terminal cell **8**. Further, like the plurality of dividing gates **10**, the purging gate **12** spans a width of the at least one aquaculture growth section **2**. In addition, the outlet weir **13** is positioned adjacent to the cell opening **4** of the second terminal cell **8** in a gravitational-higher position to allow flow of fresh water into the at least one aquaculture growth section **2** via gravity. The purging gate **12** is also slidably mounted within the second terminal cell **8**, offset to the inlet water manifold **11**, to allow the repositioning of the purging gate **12** without having to remove the purging gate **12** from the at least one aquaculture growth section **2**. Furthermore, the second terminal cell **8** is in fluid communication with the plurality of intermediate cells **9** and the first terminal cell **7** via the outlet weir **13**. In other embodiments, different fluid mechanisms can be implemented to allow the introduction of fresh water into the present invention's system.

[0058] As previously discussed, the water treatment section **14** is preferably located at one end of the at least one MCR **1**, as can be seen in FIG. **1** through **10**. However, in some embodiments, the water treatment section **14** can be located at any position in the at least one MCR **1**, as can be seen in FIG. **14** through **16**. For example, the water treatment section **14** can be positioned among the plurality of growth cells **6** of the at least one aquaculture growth section **2**. The particular chosen position can be decided based on the needs of the fish species, the specific husbandry process of the farming operation, and the specifics of the design of the existing barn. Further, the water treatment section **14s** can be designed to treat water from different aquaculture growth sections. In this embodiment, the at least one MCR **1** can include a plurality of aquaculture growth sections **40**, and each of the plurality of aquaculture growth sections **40** is in fluid communication with the water treatment section **14**.

[0059] While the system of the present invention is presented as a single integrated system, the different sections and subsystems can be implemented for separate related applications, as can be seen in FIGS. **11** and **12**. For example, the plurality of surface aerators **25** can be implemented for other aquaculture applications that are not implemented using the entirety of the present invention's system. In this embodiment, the plurality of surface aerators **25** can be operatively coupled to an aquaculture growth section, wherein each of the plurality of surface aerators **25** is used to perform some or all of the functions of biofiltration, carbon dioxide (CO₂) stripping, oxygenation, and uniform mixing and circulation of biological media to minimize dead media zones by forcing

downward flow to counterbalance the natural buoyancy of MBBR media. These functions of the plurality of surface aerators 25 are performed by the movement and aeration of system water caused by surface aerators located within the MBBR section of the specific aquaculture growth section screened contact with the media housing the treating bacteria.

[0060] Although the invention has been explained in relation to its preferred embodiment, it is to be understood that many other possible modifications and variations can be made without departing from the spirit and scope of the invention.

Claims

1. A recirculating aquaculture mixed-cell raceway system, the system comprising: at least one mixed-cell raceway; a drain system; the at least one mixed-cell raceway comprising at least one aquaculture growth section, a water treatment section, and a water return system; the water treatment section comprising a first static bed filter (SBF) reservoir, a moving bed biofilter reactor (MBBR) reservoir, and a pumping reservoir; the at least one aquaculture growth section being positioned adjacent to the water treatment section; the at least one aquaculture growth section and the water treatment section being positioned coplanar with each other; the at least one aquaculture growth section, the first SBF reservoir, the MBBR reservoir, and the pumping reservoir being in fluid communication with each other; the at least one aquaculture growth section being in fluid communication with the pumping reservoir via the drain system; and the pumping reservoir being in fluid communication with the at least one aquaculture growth section via the water return system.
2. The system as claimed in claim 1 further comprising: the at least one aquaculture growth section comprising at least one growth cell; the at least one growth cell comprising a cell opening and a cell floor; the drain system comprising at least one gravity-assisted drain, a drain conduit, and a filtration reservoir; the drain conduit being positioned external to the at least one mixed-cell raceway; the at least one gravity-assisted drain being centered on the cell floor; the at least one gravity-assisted drain being integrated into the cell floor; the at least one gravity-assisted drain being in fluid communication with the filtration reservoir via the drain conduit; and the filtration reservoir being in fluid communication with the pumping reservoir.
3. The system as claimed in claim 2, wherein the filtration reservoir is mounted within the pumping reservoir.
4. The system as claimed in claim 2 further comprising: the drain system further comprising a solids removal device; the solids removal device being mounted within the filtration reservoir; and the filtration reservoir being in fluid communication with the drain conduit via the solids removal device.
5. The system as claimed in claim 2 further comprising: the at least one growth cell being a plurality of growth cells; the at least one gravity-assisted drain being a plurality of gravity-assisted drains; the plurality of growth cells being distributed along the at least one aquaculture growth section; and each gravity-assisted drain of the plurality of growth cells being integrated into the closed base of the corresponding growth cell of the plurality of growth cells.
6. The system as claimed in claim 1 further comprising: the at least one aquaculture growth section comprising at least one growth cell; the at least one growth cell comprising a cell opening and a cell floor; the water return system comprising at least one fluid pump and at least one water return manifold; the at least one water return manifold being positioned adjacent to the cell opening; the at least one water return manifold being perimetrically connected to the at least one growth cell; the at least one fluid pump being mounted within the pumping reservoir; and the at least one water return manifold being in fluid communication with the pumping reservoir via the at least one fluid pump.
7. The system as claimed in claim 6 further comprising: the at least one growth cell being a plurality of growth cells; the at least one water return manifold being a plurality of water return

manifolds; the plurality of growth cells being distributed along the at least one aquaculture growth section; and each water return manifold of the plurality of water return manifolds being perimetrically connected to the corresponding growth cell of the plurality of growth cells.

8. The system as claimed in claim 1 further comprising: the at least one aquaculture growth section comprising at least one growth cell; the first SBF reservoir comprising a cell-to-SBF weir, an SBF-to-MBBR weir, a reservoir opening, and a reservoir floor; the cell-to-SBF weir and the SBF-to-MBBR weir being perimetrically distributed about the first SBF reservoir; the cell-to-SBF weir being positioned adjacent to the at least one growth cell; the SBF-to-MBBR weir being positioned adjacent to the MBBR reservoir; the cell-to-SBF weir and the SBF-to-MBBR weir being integrated into the reservoir opening of the first SBF reservoir; the at least one growth cell being in fluid communication with the first SBF reservoir via the cell-to-SBF weir; and the first SBF reservoir being in fluid communication with the MBBR reservoir via the SBF-to-MBBR weir.

9. The system as claimed in claim 8 further comprising: the at least one growth cell being a plurality of growth cells; the plurality of growth cells comprising a first terminal cell; the plurality of growth cells being distributed along the at least one aquaculture growth section; the first terminal cell being positioned adjacent to the water treatment section; each of the plurality of growth cells being in fluid communication with each other; and the first terminal cell being in fluid communication with the first SBF reservoir via the cell-to-SBF weir.

10. The system as claimed in claim 8 further comprising: the first SBF reservoir further comprising a SBF baffle, a SBF media section, and a SBF media barrier; the SBF baffle being mounted within the first SBF reservoir, adjacent to the cell-to-SBF weir; the SBF media barrier being mounted within the first SBF reservoir, adjacent to the reservoir opening of the first SBF reservoir; and the SBF media section being positioned in between the SBF media barrier and the SBF baffle.

11. The system as claimed in claim 1 further comprising: the water treatment section further comprising a second SBF reservoir; and the at least one aquaculture growth section, the first SBF reservoir, the MBBR reservoir, the second SBF reservoir, and the pumping reservoir being in fluid communication with each other.

12. The system as claimed in claim 11 further comprising: the second SBF reservoir comprising a MBBR-to-SBF weir, an SBF-to-pump weir, a reservoir opening, and a reservoir floor; the MBBR-to-SBF weir and the SBF-to-pump weir being perimetrically distributed about the second SBF reservoir; the MBBR-to-SBF weir being positioned adjacent to the MBBR reservoir; the SBF-to-pump weir being positioned adjacent to the pumping reservoir; the MBBR-to-SBF weir and the SBF-to-pump weir being integrated into the reservoir opening of the second SBF reservoir; the MBBR reservoir being in fluid communication with the second SBF reservoir via the MBBR-to-SBF weir; and the second SBF reservoir being in fluid communication with the pumping reservoir via the SBF-to-pump weir.

13. The system as claimed in claim 1 further comprising: the MBBR reservoir comprising a MBBR-to-pump weir and a reservoir opening; the MBBR-to-pump weir being perimetrically distributed about the MBBR reservoir; the MBBR-to-pump weir being positioned adjacent to the pumping reservoir; the MBBR-to-pump weir being integrated into the reservoir opening of the MBBR reservoir; and the pumping reservoir being in fluid communication with the MBBR reservoir via the MBBR-to-pump weir.

14. The system as claimed in claim 1 further comprising: the MBBR reservoir comprising a plurality of surface aerators; the plurality of surface aerators being distributed throughout the MBBR reservoir; and the plurality of surface aerators being mounted within the MBBR reservoir.

15. The system as claimed in claim 1 further comprising: the at least one aquaculture growth section comprising a plurality of growth cells; the plurality of growth cells comprising a first terminal cell, a second terminal cell, and a plurality of intermediate cells; the plurality of growth cells being distributed along the at least one aquaculture growth section; the first terminal cell being positioned adjacent to the water treatment section; the second terminal cell being positioned

opposite the first terminal cell along the at least one aquaculture growth section; and the plurality of intermediate cells being positioned in between the first terminal cell and the second terminal cell.

16. The system as claimed in claim 15 further comprising: the at least one aquaculture growth section further comprising a plurality of dividing gates; each of the plurality of dividing gates spanning a width of the at least one aquaculture growth section; a dividing gate of the plurality of dividing gates being positioned in between the first terminal cell and an adjacent intermediate cell of the plurality of intermediate cells; a dividing gate of the plurality of dividing gates being positioned in between the second terminal cell and an adjacent intermediate cell of the plurality of intermediate cells; a dividing gate of the plurality of dividing gates being positioned in between an arbitrary intermediate cell and an adjacent intermediate cell of the plurality of intermediate cells; and each of the plurality of dividing gates being slidably mounted within the at least one aquaculture growth section.

17. The system as claimed in claim 15, wherein each dividing gate of the plurality of dividing gates is made of a water permeable material.

18. The system as claimed in claim 15 further comprising: the at least one aquaculture growth section further comprising an inlet water manifold and a purging gate; the purging gate comprising an outlet weir; the inlet water manifold being positioned adjacent to the second terminal cell; the inlet water manifold being mounted onto a cell opening of the second terminal cell; the purging gate spanning a width of the at least one aquaculture growth section; the outlet weir being positioned adjacent to the cell opening of the second terminal cell; the purging gate being slidably mounted within the second terminal cell, offset to the inlet water manifold; and the second terminal cell being in fluid communication with the plurality of intermediate cells and the first terminal cell via the outlet weir.

19. The system as claimed in claim 1 further comprising: the at least one aquaculture growth section being a plurality of aquaculture growth sections; and each of the plurality of aquaculture growth sections being in fluid communication with the water treatment section.

20. A moving bed bio reactor (MBBR) system, the system comprising: a water vessel; one or more surface aerators; a screening mechanism; an amount of biological media; and wherein the screening mechanism is used to prevent the amount of biological media from entering the one or more surface aerators, and wherein the one or more surface aerators are used to perform some or all of the functions of biofiltration, carbon dioxide (CO₂) stripping, oxygenation, and uniform mixing and circulation of the amount of biological media to minimize dead media zones by forcing downward flow to counterbalance the natural buoyancy of MBBR media.
